

Federated Training

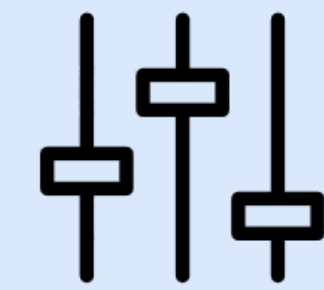
server

$$\hat{W}_i = \frac{N_i \cdot (W_i + W_j)}{N_i + N_j}$$

N_i, W_i

N_i, W_i

$$\hat{W}_j = \frac{N_j \cdot (W_i + W_j)}{N_i + N_j}$$



Backbone $\triangleright$ Head

Adaptation rates α

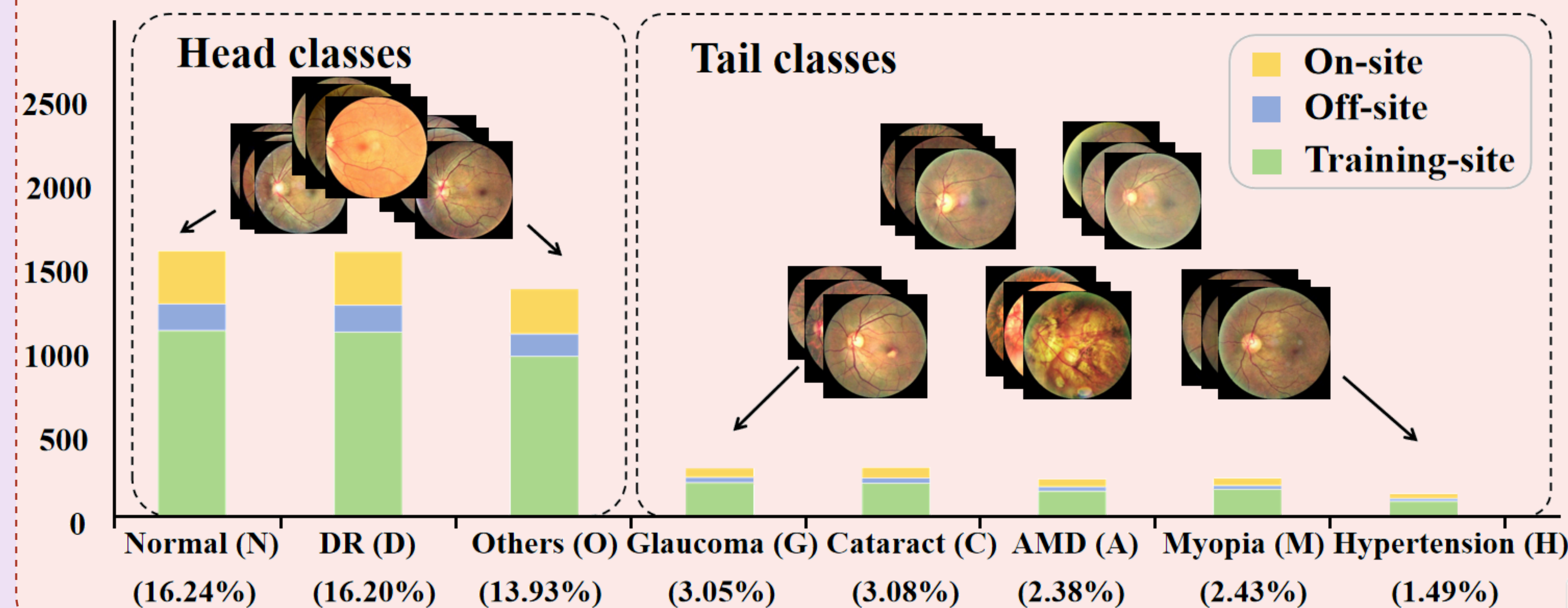
Loss

preprocessing

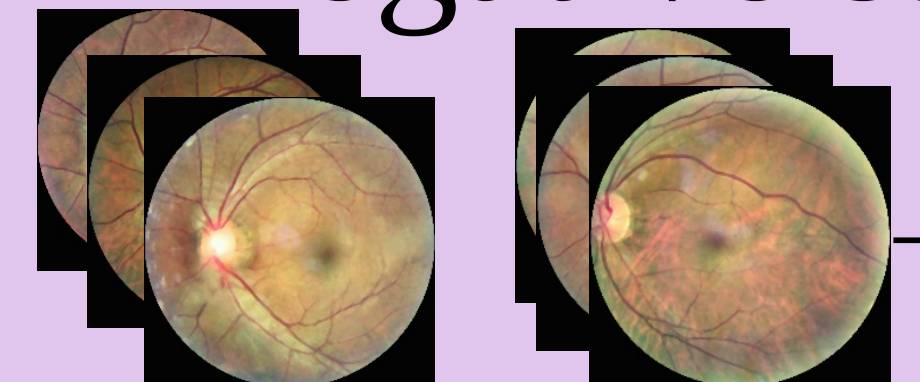
Head classes



Long-tail classes

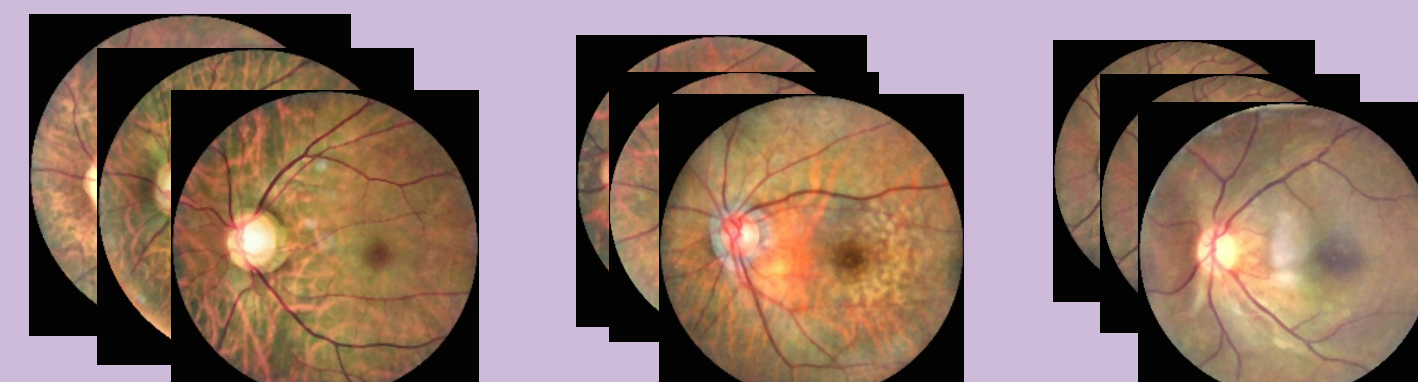


Negative sample loss



Hill regularization

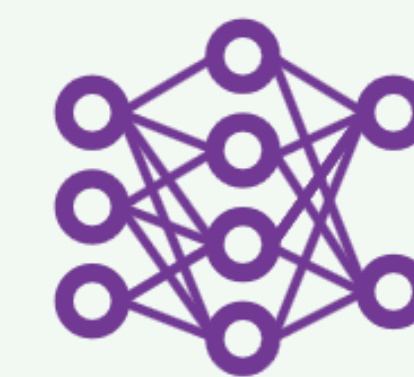
Positive samples loss



Loss aggregation

Calibration (fit on validation)

Global model



Fit calibration

Optimizer

Objective

T, b, W

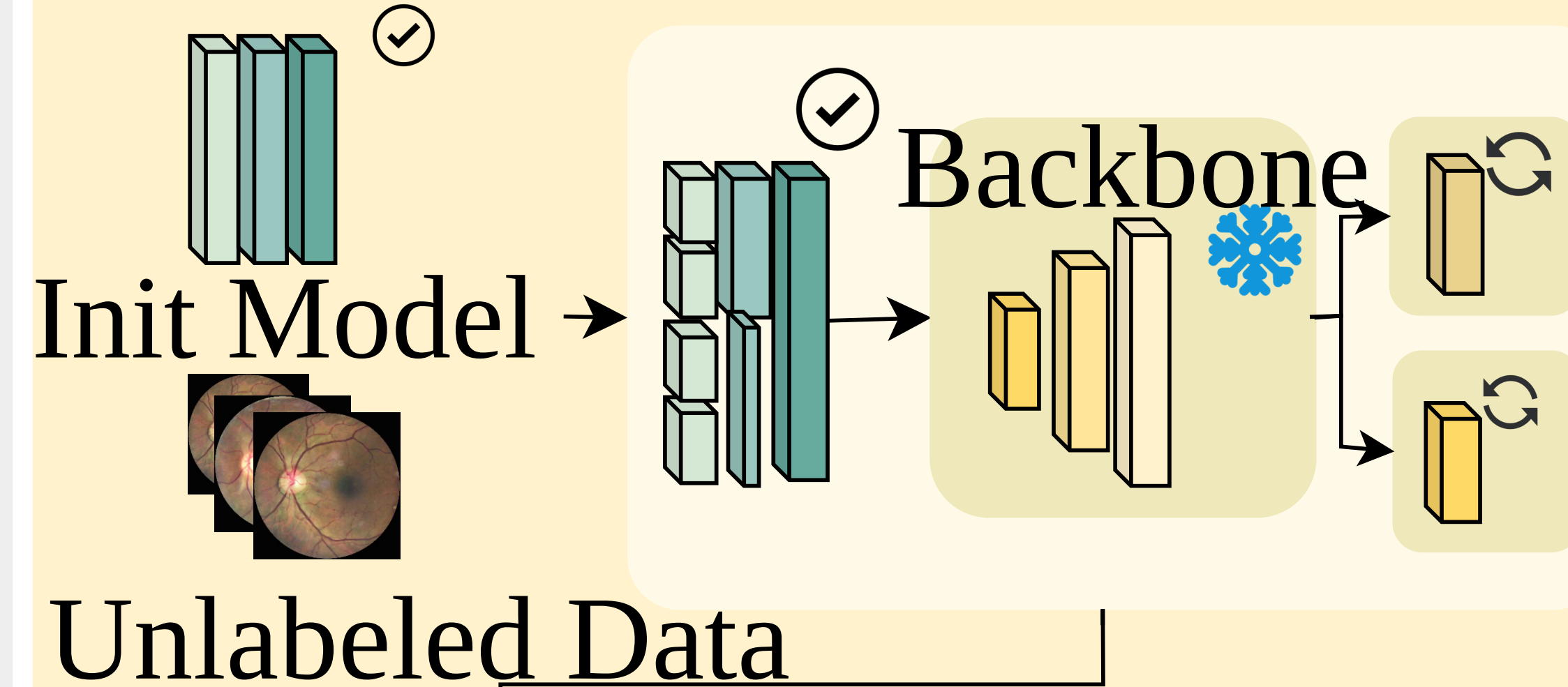
Calibration mapping $\mathcal{C}$
($z; T, b, W$)

Estimated source prior $\hat{p}(y)$

save (T, b, W) for test-time



Personalization



batch

online avg

Personalized Prediction

Upload

Download

Calibration Mapping

Exponential Moving Average update

Calibrated Model

BN stats
BN γ/β

Linear W/b

T
Logit temperature

b
Class-wise bias

W
Linear mapping for prior correction